Projection Learning and Conservative Projection Memory

Anonymous

Abstract

We introduce Projection Learning, a variational framework for runtime learning in which state updates are minimal consistency repairs. The update is a Bregman projection onto an evidence set, or its soft proximal analogue. We prove that projection is characterized by a Pythagorean principle: it is exactly the update that satisfies evidence while not increasing distance to any feasible ideal state. We also show that soft Projection Learning is implicit mirror descent, with ordinary gradient descent recovered as a linearized Euclidean special case. We instantiate the framework as Conservative Projection Memory, a neural associative state whose write direction is the innovation of a key relative to previously protected evidence. This yields an algebraic decomposition into redundant, novel, and contradictory evidence. We prove exact binding and capacity laws, conflict impossibility, no-free-repair lower bounds, optimal residual contraction, equivalence between sequential hard updates and batch projection, ridge competitive bounds, log-determinant information growth, and Bayesian interpretations. The resulting algorithms perform runtime adaptation by forward computation plus closed-form state repair, without inference-time backpropagation.

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1 Introduction

Backpropagation made hidden representations trainable by turning learning into differentiation through a static computation graph. This is the right primitive when the object being optimized is a parameterized function and the task is to distribute a scalar error through differentiable structure. Runtime learning asks for a different primitive. A deployed system receives evidence after it has already formed state, and the update must incorporate that evidence without changing what the evidence does not touch.

The central problem is therefore not only optimization but controlled repair. If the current state is s_t and the evidence at time t determines a set of acceptable states $\mathcal{C}_t$, the natural update is

$$s_{t+1} = \operatorname*{argmin}_{s \in \mathcal{C}_t} D_\psi(s, s_t).$$

If the evidence is noisy, the corresponding soft update is

$$s_{t+1} = \operatorname*{argmin}_{s} \{D_\psi(s, s_t) + \eta_t \ell_t(s)\}.$$

This paper studies these updates as a runtime learning principle. We call the principle Projection Learning: learning is minimal consistency repair.

The first contribution is a characterization theorem. Bregman projection is not merely one possible update. It is exactly the update that satisfies the evidence and obeys a Pythagorean inequality against every feasible ideal state. Thus the variational rule is equivalent to a stability statement: after projection, the learner has not moved away from any state that also satisfies the evidence.

The second contribution is a canonical neural realization called Conservative Projection Memory. A runtime memory is a matrix $W_t \in \mathbb{R}^{m \times r}$, and evidence is a key-value constraint $W k_t = v_t$. If old keys form K_{t-1}, the correct write direction is not k_t but the innovation

$$z_t = (I - \Pi_{t-1}) k_t, \qquad \Pi_{t-1} = K_{t-1} K_{t-1}^{+}.$$

The soft update is

$$W_{t+1} = W_t + \frac{\eta_t}{1 + \eta_t \|z_t\|^2} (v_t - W_t k_t) z_t^\top.$$

This is the smallest update that repairs the new residual while preserving old constraints whenever such preservation is possible.

The innovation view gives the mechanism its semantics. If $\|z_t\|$ is large, the key contains new information and the state can be updated in a new direction. If $z_t = 0$ and the implied value agrees with the evidence, the observation is redundant. If $z_t = 0$ and the value disagrees, the evidence is contradictory: no update can satisfy it while preserving the old bindings. The method therefore exposes novelty and conflict as algebraic facts, not as hidden consequences of an optimizer.

Organization. Section 2 fixes notation and the projection geometry. Section 3 states the main informal results. Section 4 gives executable algorithms and complexity. Section 5 gives proof sketches. Section 6 concludes. Appendix A gives a guide to the appendix; Appendix B states limitations; Appendix C discusses related work; the remaining appendices contain formal statements and proofs.

2 Preliminaries

Let S be a closed convex subset of a finite-dimensional real vector space. Let $\psi : S \to \mathbb{R}$ be differentiable and strictly convex. The Bregman divergence generated by ψ is

$$D_\psi(u, v) = \psi(u) - \psi(v) - \langle \nabla\psi(v), u - v \rangle .$$

When $\psi(s) = \|s\|_2^2/2$, this is Euclidean squared distance. All projection statements below are finite-dimensional and use the standard variational characterization of convex minimizers.

For a matrix K, K^+ denotes the Moore-Penrose pseudoinverse, $\ker K$ its nullspace, and KK^+ the orthogonal projector onto the column span of K. A memory state is a matrix $W \in \mathbb{R}^{m\times r}$. A collection of key-value evidence is written as

$$K = [k_1, \dots, k_T] \in \mathbb{R}^{r\times T}, \qquad V = [v_1, \dots, v_T] \in \mathbb{R}^{m\times T}.$$

Exact binding means $WK = V$. A new key k is decomposed relative to the old key span by

$$\Pi = KK^+, \qquad z = (I - \Pi)k, \qquad a = K^+k. \tag{1}$$

The vector z is the innovation. It is the component of k not explained by previous keys.

The paper uses two levels of theorem statements. Main-text theorems are informal but mathematically precise enough to communicate the result and proof method. Each such theorem points to a formal appendix theorem. The appendix theorem links back to its main-text version.

3 Main Results

3.1 Projection as the canonical repair update

Theorem 3.1 (Projection characterization, informal version of Theorem D.2)**.** *Let $\mathcal{C} \subseteq S$ be closed and convex. A point $p \in \mathcal{C}$ is the Bregman projection of s_0 onto $\mathcal{C}$ if and only if, for every $u \in \mathcal{C}$,*

$$D_\psi(u, p) + D_\psi(p, s_0) \le D_\psi(u, s_0).$$

This says that minimal repair and global stability with respect to all feasible ideal states are the same condition.

3.2 Relation to gradient learning

Theorem 3.2 (Soft projection as implicit mirror descent, informal version of Theorem D.5)**.** *Let ℓ_t be differentiable and suppose the soft projection optimum is interior:*

$$s_{t+1} = \operatorname*{argmin}_s \{D_\psi(s, s_t) + \eta\ell_t(s)\}.$$

Then

$$\nabla\psi(s_{t+1}) = \nabla\psi(s_t) - \eta\nabla\ell_t(s_{t+1}).$$

For $\psi(s) = \|s\|_2^2/2$, this becomes implicit gradient descent. Replacing $\nabla\ell_t(s_{t+1})$ by $\nabla\ell_t(s_t)$ gives ordinary explicit gradient descent.

Theorem 3.3 (Quadratic projection memory as exact implicit gradient descent, informal version of Theorem D.6)**.** *Let* $\ell(W) = \|Wk - v\|^2/2$*. The Euclidean soft projection update*

$$W_{t+1} = \operatorname*{argmin}_{W} \left\{ \frac{1}{2} \|W - W_t\|_F^2 + \eta \ell(W) \right\}$$

is the implicit gradient step $W_{t+1} = W_t - \eta \nabla \ell(W_{t+1})$ *and has the closed form*

$$W_{t+1} = W_t + \frac{\eta}{1 + \eta \|k\|^2} (v - W_t k) k^\top.$$

Its residual contracts by $(1 + \eta \|k\|^2)^{-1}$ *for all* $\eta > 0$*.*

3.3 Exact binding and capacity

Theorem 3.4 (Exact associative binding, informal version of Theorem E.1)**.** *For key-value matrices* $K \in \mathbb{R}^{r \times T}$ *and* $V \in \mathbb{R}^{m \times T}$*, the constraint set* $\{W : WK = V\}$ *is nonempty if and only if* $\ker K \subseteq \ker V$*. When it is nonempty, the Euclidean projection of* W_0 *onto this set is*

$$W^\star = W_0 + (V - W_0 K) K^+.$$

In particular, if K *has full column rank, then arbitrary values can be exactly bound.*

Thus an r-dimensional key space exactly stores arbitrary values for at most r independent keys. Beyond that rank, values must obey the same linear relations as keys.

3.4 Innovation, conflict, and conservative writing

Theorem 3.5 (Innovation decomposition, informal version of Theorem E.2)**.** *Assume old evidence is* $WK = V$ *and decompose a new key as in* (1)*. Every old-feasible memory satisfies*

$$Wk = Va + Wz.$$

If $z = 0$*, the value at* k *is already fixed by old evidence. If* $z \neq 0$*, the new evidence constrains only the innovation direction through* $Wz = v - Va$*.*

Theorem 3.6 (Conflict impossibility, informal version of Theorem E.3)**.** *If* $z = 0$*, then no update preserving* $WK = V$ *can change* Wk*. Hence evidence* $Wk = v$ *is compatible with old evidence if and only if* $v = W_t k$*.*

Theorem 3.7 (Soft Conservative Projection Memory, informal version of Theorem E.8)**.** *Assume* $W_t K = V$ *and* $z \neq 0$*. The constrained soft repair problem*

$$W_{t+1} = \operatorname*{argmin}_{W : WK = V} \left\{ \frac{1}{2} \|W - W_t\|_F^2 + \frac{\eta}{2} \|Wk - v\|^2 \right\}$$

has the closed form

$$W_{t+1} = W_t + \frac{\eta}{1 + \eta \|z\|^2} (v - W_t k) z^\top.$$

It preserves old constraints and contracts the new residual by the factor $(1 + \eta \|z\|^2)^{-1}$*.*

Theorem 3.8 (No-free-repair lower bound, informal version of Theorem E.5)**.** *Assume $W_t K = V$, $z = (I - \Pi)k$, and $e = v - W_t k$. Let an update $W_t + \Delta$ preserve old constraints, so $\Delta K = 0$. If the post-update residual obeys*

$$\|v - (W_t + \Delta)k\| \leq \gamma \|e\|$$

for $0 \leq \gamma < 1$ and $z \neq 0$, then

$$\|\Delta\|_F \geq \frac{(1-\gamma)\|e\|}{\|z\|}.$$

If $z = 0$ and $e \neq 0$, no such contraction is possible while preserving old constraints.

Theorem 3.9 (Optimal contraction, informal version of Theorem E.6)**.** *Fix $0 \leq \gamma \leq 1$ and $z \neq 0$. Among all updates preserving $\Delta K = 0$ and achieving exact residual $v - (W_t + \Delta)k = \gamma e$, the minimum-Frobenius-norm update is*

$$\Delta^\star = (1-\gamma)\frac{ez^\top}{\|z\|^2}.$$

It attains the lower bound in Theorem 3.8. Soft CPM is the special case $\gamma = (1 + \eta\|z\|^2)^{-1}$.

Theorem 3.10 (Sequential hard CPM equals batch projection, informal version of Theorem E.7)**.** *For compatible evidence $(k_t, v_t)_{t=1}^T$, sequential hard CPM with redundant steps skipped returns*

$$W_{T+1} = \operatorname*{argmin}_{W: WK_T = V_T} \frac{1}{2}\|W - W_0\|_F^2 = W_0 + (V_T - W_0 K_T)K_T^+.$$

Thus the online hard update exactly tracks the batch Euclidean projection solution.

3.5 Ridge and Bayesian memory

Theorem 3.11 (Ridge and Bayesian projection memory, informal version of Theorem F.5)**.** *The ridge objective*

$$J_t(W) = \frac{\lambda}{2}\|W - W_0\|_F^2 + \frac{1}{2}\sum_{\tau=1}^{t} \rho_\tau \|Wk_\tau - v_\tau\|^2$$

has minimizer $W_{t+1} = B_t A_t^{-1}$, where

$$A_t = \lambda I + \sum_{\tau=1}^{t} \rho_\tau k_\tau k_\tau^\top, \qquad B_t = \lambda W_0 + \sum_{\tau=1}^{t} \rho_\tau v_\tau k_\tau^\top.$$

The same matrix is the posterior mean and MAP estimate under a Gaussian linear observation model.

Theorem 3.12 (Ridge bound, informal version of Theorem F.3)**.** *Let the pre-update ridge state and covariance be*

$$W_t = \operatorname*{argmin}_W J_{t-1}(W), \qquad P_t = \left(\lambda I + \sum_{\tau<t} \rho_\tau k_\tau k_\tau^\top\right)^{-1}.$$

Define $a_t = k_t^\top P_t k_t$. For every comparator U,

$$\sum_{t=1}^{T} \frac{\rho_t}{1 + \rho_t a_t}\|v_t - W_t k_t\|^2 \leq \lambda\|U - W_0\|_F^2 + \sum_{t=1}^{T} \rho_t \|v_t - Uk_t\|^2.$$

Theorem 3.13 (Log-determinant information identity, informal version of Theorem F.4)**.** *Let* $A_t = \lambda I + \sum_{\tau=1}^{t} \rho_\tau k_\tau k_\tau^\top$ *and* $a_t = k_t^\top A_{t-1}^{-1} k_t$*. Then*

$$\sum_{t=1}^{T} \log(1 + \rho_t a_t) = \log \frac{\det A_T}{\det A_0}.$$

4 Algorithms and Complexity

Projection memory separates slow trainable parameters from fast runtime state. The slow parameters are the backbone and the maps K, V, B. They are trained by ordinary backpropagation. The fast state is updated by projection both during training and inference.

4.1 Streaming Conservative Projection Memory

The exact conservative algorithm maintains $W_t \in \mathbb{R}^{m \times r}$ and an orthonormal basis $Q_t \in \mathbb{R}^{r \times q_t}$ for protected key directions.

- **Part 1. Read.** Given $h_t \in \mathbb{R}^d$, compute

$$k_t = Kh_t, \qquad v_t = Vh_t, \qquad r_t = W_t k_t, \qquad \tilde{h}_t = h_t + Br_t.$$

- **Part 2. Diagnose.** Compute

$$z_t = k_t - Q_t Q_t^\top k_t, \qquad e_t = v_t - W_t k_t.$$

- **Part 3. Write.** If $\|z_t\|^2 > \varepsilon$, update

$$W_{t+1} = W_t + \frac{\eta_t}{1 + \eta_t \|z_t\|^2} e_t z_t^\top$$

  and optionally append $z_t / \|z_t\|$ to Q_t. If $\|z_t\|^2 \le \varepsilon$ and $\|e_t\|$ is small, the evidence is redundant. If $\|z_t\|^2 \le \varepsilon$ and $\|e_t\|$ is large, the evidence is conflicting relative to protected memory.

The per-step overhead is

$$O(d(r + m) + dm + mr + rq_t).$$

If $m = r$ and $q_t \le r$, this is

$$O(dr + mr + r^2).$$

The runtime state cost is

$$O(mr + rq_t).$$

4.2 Ridge Projection Memory

The ridge algorithm maintains $W_t \in \mathbb{R}^{m\times r}$ and $P_t \in \mathbb{R}^{r\times r}$, initialized by

$$W_1 = W_0, \qquad P_1 = \lambda^{-1} I.$$

At step t, after computing $k_t, v_t, r_t, \tilde{h}_t$, choose $\rho_t \geq 0$ and update

$$\begin{aligned} g_t &= \frac{\rho_t P_t k_t}{1 + \rho_t k_t^\top P_t k_t}, \\ W_{t+1} &= W_t + (v_t - W_t k_t) g_t^\top, \\ P_{t+1} &= P_t - \frac{\rho_t P_t k_t k_t^\top P_t}{1 + \rho_t k_t^\top P_t k_t}. \end{aligned}$$

The per-step overhead is

$$O(d(r+m) + dm + mr + r^2),$$

and the runtime state cost is

$$O(mr + r^2).$$

4.3 Training and inference

The same state update is used in training and inference.

- **Part 1. Fully unrolled training.** Projection updates are unrolled and differentiated through. This trains K, V, B for long-horizon memory effects but stores runtime states across the sequence.
- **Part 2. Truncated state-gradient training.** Gradients are propagated through only the most recent H state updates, giving state memory $O(HM_s)$ for runtime-state size M_s.
- **Part 3. Detached-state training.** Projection updates are executed with stop-gradient. This has inference-like memory cost, but long-horizon credit to V may require auxiliary consistency losses or periodic unrolling.

At inference, there is no backward pass. The cost per token is

$$O(C_{\text{bb}} + C_{\text{pm}}),$$

where C_{bb} is the backbone forward cost and C_{pm} is the projection-memory overhead above. In contrast, TTT-style adaptation typically requires an inner backward pass through the test-time module. Projection memory is forward computation plus closed-form rank-one state repair.

5 Technical Overview

This section sketches the proof ideas. Full formal statements and proofs appear in Appendices D, E, and F.

Theorem 5.1 (Projection characterization, proof sketch for Theorem D.2)**.** *The informal statement is Theorem 3.1.*

Proof sketch. The optimality condition for $p = \operatorname{argmin}_{s \in \mathcal{C}} D_\psi(s, s_0)$ is

$$\langle \nabla\psi(p) - \nabla\psi(s_0), u - p \rangle \geq 0 \qquad \forall u \in \mathcal{C}.$$

The Bregman three-point identity gives

$$D_\psi(u, s_0) - D_\psi(u, p) - D_\psi(p, s_0) = \langle \nabla\psi(s_0) - \nabla\psi(p), p - u \rangle .$$

The two displays are equivalent. This proves both directions. □

Theorem 5.2 (Implicit mirror descent, proof sketch for Theorem D.5)**.** *The informal statement is Theorem 3.2.*

Proof sketch. Differentiate the soft projection objective at the interior optimum:

$$0 = \nabla_s D_\psi(s_{t+1}, s_t) + \eta \nabla \ell_t(s_{t+1}).$$

Since

$$\nabla_s D_\psi(s, s_t) = \nabla\psi(s) - \nabla\psi(s_t),$$

we obtain

$$\nabla\psi(s_{t+1}) = \nabla\psi(s_t) - \eta \nabla \ell_t(s_{t+1}).$$

□

Theorem 5.3 (Quadratic memory and implicit gradient descent, proof sketch for Theorem D.6)**.** *The informal statement is Theorem 3.3.*

Proof sketch. For $\ell(W) = \|Wk - v\|^2 / 2$,

$$\nabla \ell(W) = (Wk - v)k^\top.$$

The implicit step is

$$W_{t+1} = W_t + \eta(v - W_{t+1}k)k^\top.$$

With $e = v - W_t k$ and $W_{t+1} = W_t + \alpha e k^\top$,

$$v - W_{t+1}k = (1 - \alpha \|k\|^2)e.$$

Comparing coefficients gives

$$\alpha = \eta(1 - \alpha \|k\|^2) = \frac{\eta}{1 + \eta \|k\|^2}.$$

□

Theorem 5.4 (Exact binding, proof sketch for Theorem E.1)**.** *The informal statement is Theorem 3.4.*

Proof sketch. If $WK = V$ and $Ka = 0$, then

$$Va = WKa = 0.$$

Thus $\ker K \subseteq \ker V$. Conversely this nullspace condition implies

$$V = VK^+K,$$

so $W = VK^+$ is feasible. For projection, write $W = W_0 + \Delta$ and $R = V - W_0K$. The problem becomes

$$\min_{\Delta} \frac{1}{2} \|\Delta\|_F^2 \quad \text{subject to} \quad \Delta K = R.$$

The minimum-norm feasible solution is $\Delta = RK^+$. □

Theorem 5.5 (Innovation and conflict, proof sketch for Theorems E.2 and E.3)**.** *The informal statements are Theorems 3.5 and 3.6.*

Proof sketch. Using (1),

$$k = KK^+k + (I - KK^+)k = Ka + z.$$

If $WK = V$, then

$$Wk = WKa + Wz = Va + Wz.$$

If $z = 0$, then $Wk = Va$ for every old-feasible W. If an update Δ preserves old evidence, then $\Delta K = 0$ and hence $\Delta k = 0$ whenever $k \in \text{span}(K)$. Thus contradiction cannot be repaired without breaking old evidence. □

Theorem 5.6 (Soft CPM, proof sketch for Theorem E.8)**.** *The informal statement is Theorem 3.7.*

Proof sketch. Set $W = W_t + \Delta$ and $e = v - W_tk$. Since $W_tK = V$, preserving old evidence is $\Delta K = 0$. Because $\Delta K = 0$,

$$\Delta k = \Delta(I - \Pi)k = \Delta z.$$

The objective becomes

$$\min_{\Delta:\Delta K=0} \frac{1}{2} \|\Delta\|_F^2 + \frac{\eta}{2} \|\Delta z - e\|^2 .$$

The stationarity equation inside this subspace is

$$\Delta + \eta(\Delta z - e)z^\top = 0.$$

Solving with $\Delta = \alpha e z^\top$ yields

$$\alpha = \frac{\eta}{1 + \eta \|z\|^2}.$$

The residual satisfies

$$v - W_{t+1}k = e - \Delta z = \frac{1}{1 + \eta \|z\|^2} e.$$

□

Theorem 5.7 (No-free repair and optimal contraction, proof sketch for Theorems E.5 and E.6)**.** *The informal statements are Theorems 3.8 and 3.9.*

Proof sketch. Every old-preserving update satisfies

$$\Delta k = \Delta(I - \Pi)k = \Delta z.$$

The post-update residual is

$$e^+ = e - \Delta z.$$

If $\|e^+\| \le \gamma \|e\|$, then

$$\|\Delta z\| = \left\|e - e^+\right\| \ge (1-\gamma)\|e\|.$$

Since $\|\Delta z\| \le \|\Delta\|_F \|z\|$, the lower bound follows. If one demands $e^+ = \gamma e$, then $\Delta z = (1-\gamma)e$, and the minimum-norm solution is

$$\Delta^\star = (1-\gamma)\frac{ez^\top}{\|z\|^2}.$$

□

Theorem 5.8 (Sequential hard CPM, proof sketch for Theorem E.7)**.** *The informal statement is Theorem 3.10.*

Proof sketch. Let $\mathcal{C}_t = \{W : WK_t = V_t\}$ and $\mathcal{T}_t = \{H : HK_t = 0\}$. By induction, $W_t \in \mathcal{C}_{t-1}$ and

$$\langle H, W_t - W_0\rangle_F = 0 \qquad \forall H \in \mathcal{T}_{t-1}.$$

The hard CPM update lies in $\mathcal{C}_t$. If $H \in \mathcal{T}_t$, then $Hz_t = 0$, so

$$\left\langle H, e_t z_t^\top \right\rangle_F = e_t^\top H z_t = 0.$$

Thus $W_{t+1} - W_0$ is orthogonal to $\mathcal{T}_t$, which is the Euclidean projection condition for $\mathcal{C}_t$. □

Theorem 5.9 (Ridge and Bayesian memory, proof sketch for Theorem F.5)**.** *The informal statement is Theorem 3.11.*

Proof sketch. Differentiating J_t gives

$$0 = \lambda(W - W_0) + \sum_{\tau=1}^{t} \rho_\tau (Wk_\tau - v_\tau)k_\tau^\top.$$

Rearranging,

$$W\left(\lambda I + \sum_{\tau=1}^{t} \rho_\tau k_\tau k_\tau^\top\right) = \lambda W_0 + \sum_{\tau=1}^{t} \rho_\tau v_\tau k_\tau^\top.$$

Thus $W = B_t A_t^{-1}$. Under the Gaussian prior and observation model, the negative log posterior equals $J_t(W)$ up to constants, so the posterior mean and MAP coincide with this minimizer. □

Theorem 5.10 (Ridge competitive bound and log determinant, proof sketch for Theorems F.3 and F.4)**.** *The informal statements are Theorems 3.12 and 3.13.*

Proof sketch. Around the minimizer W_t of J_{t-1}, the increment minimization is

$$\min_{\Delta} \frac{1}{2} \operatorname{tr}(\Delta A_{t-1} \Delta^\top) + \frac{\rho_t}{2} \|\Delta k_t - e_t\|^2 .$$

Its minimum value is

$$\frac{\rho_t}{2(1+\rho_t a_t)} \|e_t\|^2 .$$

Summing and comparing $J_T(W_{T+1})$ with $J_T(U)$ gives the competitive bound. The log-determinant identity follows from

$$\det(A_{t-1} + \rho_t k_t k_t^\top) = \det(A_{t-1})(1 + \rho_t a_t).$$

□

6 Conclusion

Projection Learning treats runtime learning as minimal consistency repair. Its primitive is projection of state under evidence, with gradient descent recovered as a linearized Euclidean special case. Conservative Projection Memory realizes this principle in a neural associative state by writing along the innovation of a key relative to protected evidence. This yields feasibility conditions, capacity laws, conflict detection, lower bounds on repair, optimal residual contraction, online equivalence to batch projection, and ridge competitive guarantees. The main conceptual conclusion is that runtime learning should expose the geometry of evidence: what is already implied, what is genuinely new, and what is inconsistent with what the system has committed to preserve.

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Appendix

A Appendix Roadmap

The appendix is organized to mirror the logic of the paper while giving full formal details.

- **Part 1.** Appendix B states the limitations of the framework and clarifies what is not proved.
- **Part 2.** Appendix C gives related work after the technical story, so the main text can stay focused on definitions and results.
- **Part 3.** Appendix D proves the projection-learning facts: the Bregman identity, the projection characterization theorem, soft projection bounds, KL projection, and the gradient-descent relation.
- **Part 4.** Appendix E proves the exact memory theory: feasibility, capacity, innovation decomposition, conflict impossibility, no-free-repair lower bounds, optimal contraction, sequential equivalence to batch projection, and hard and soft conservative updates.
- **Part 5.** Appendix F proves the ridge-recursive update, Bayesian interpretation, normalized competitive bound, and log-determinant information identity.

B Limitations

Projection Learning is a formal update principle, not a guarantee that every source of evidence is useful. If the value map Vh_t produces a poor target, projection will repair the wrong constraint. The theory also separates feasibility from generalization: exact binding says when constraints can be satisfied, but it does not imply that the resulting memory improves downstream prediction.

Conservative Projection Memory assumes that preserving old constraints is desirable. This is appropriate for stable binding, but nonstationary environments may require forgetting, reset, branching, or discounted ridge objectives. Exact innovation maintenance also has a computational cost. In large systems, one may use rank caps, approximate orthogonalization, diagonal or low-rank covariance estimates, or learned gates; these approximations are not covered by the exact preservation theorems.

The paper proves deterministic algebraic, convex-analytic, and online regression statements. It does not prove end-to-end performance improvements for language modeling, classification under shift, or long-context retrieval. Those claims require experiments. The proposed diagnostics are residual, innovation norm, conflict rate, leverage, and contraction ratio; the theory predicts their behavior under the stated model, not their empirical distribution in trained networks.

C Related Work

Backpropagation and runtime updates. Backpropagation supplies differentiable credit assignment for static computation graphs [RHW86]. Synthetic gradients decouple modules by predicting downstream gradient information [JCO+17]. Projection Learning decouples runtime updates differently: the update is not a predicted derivative, but the solution of a local state-repair problem.

Bregman projection and online optimization. Bregman projections originate in convex feasibility and optimization [Bre67]. Online convex optimization, mirror descent, and proximal methods use related geometries to bound regret and movement [Zin03, SS12, Haz16]. The present paper uses this geometry as a runtime-learning semantics: evidence defines consistency, and projection defines the least disruptive repair. The implicit mirror-descent view clarifies that gradient descent is not rejected, but recovered as a linearized Euclidean special case.

Passive-aggressive learning and least squares. Passive-aggressive algorithms solve minimal-change constrained updates for supervised online learning [CDK+06]. Recursive least squares and online ridge regression yield classical closed-form updates and loss bounds [Vov97, AW01]. Conservative Projection Memory adapts these ideas to internal neural state and adds the innovation/conflict decomposition induced by old evidence.

Test-time training and expressive hidden states. Test-time training updates model parameters on self-supervised objectives at inference time [SWL+20]. TTT layers make hidden states themselves trainable models updated on test sequences [SLD+25]. Conservative Projection Memory targets the same need for runtime adaptation, but replaces test-time backpropagation with closed-form projection updates.

Fast weights and linear attention. Fast weights store temporary information in rapidly changing associative matrices [BHM+16]. Linear attention can be interpreted through recurrent fast-weight updates [KVPF20, SIS21]. Conservative Projection Memory differs by writing residuals rather than raw values and by writing in the key innovation subspace rather than along the raw key.

D Projection Learning Proofs

Lemma D.1 (Bregman three-point identity)**.** *For all $a, b, u \in S$,*

$$D_\psi(u, a) - D_\psi(u, b) - D_\psi(b, a) = \langle \nabla\psi(a) - \nabla\psi(b), b - u \rangle .$$

Proof. Expand each term:

$$\begin{aligned} D_\psi(u, a) &= \psi(u) - \psi(a) - \langle \nabla\psi(a), u - a \rangle , \\ D_\psi(u, b) &= \psi(u) - \psi(b) - \langle \nabla\psi(b), u - b \rangle , \\ D_\psi(b, a) &= \psi(b) - \psi(a) - \langle \nabla\psi(a), b - a \rangle . \end{aligned}$$

Then

$$\begin{aligned} D_\psi(u, a) - D_\psi(u, b) - D_\psi(b, a) &= - \langle \nabla\psi(a), u - a \rangle + \langle \nabla\psi(b), u - b \rangle + \langle \nabla\psi(a), b - a \rangle \\ &= \langle \nabla\psi(a), b - u \rangle - \langle \nabla\psi(b), b - u \rangle \\ &= \langle \nabla\psi(a) - \nabla\psi(b), b - u \rangle . \end{aligned}$$

□

Theorem D.2 (Projection characterization, formal version of Theorem 3.1)**.** *Let S be closed and convex, let ψ be differentiable and strictly convex, let $\mathcal{C} \subseteq S$ be closed and convex, and let $s_0 \in S$. For $p \in \mathcal{C}$, the following are equivalent:*

- **Part 1.** $p = \operatorname{argmin}_{s \in \mathcal{C}} D_\psi(s, s_0)$.
- **Part 2.** *For all* $u \in \mathcal{C}$,

$$D_\psi(u, p) + D_\psi(p, s_0) \leq D_\psi(u, s_0).$$

Proof. Assume Part 1. The first-order optimality condition is

$$\langle \nabla\psi(p) - \nabla\psi(s_0), u - p \rangle \geq 0 \qquad \forall u \in \mathcal{C}.$$

Equivalently,

$$\langle \nabla\psi(s_0) - \nabla\psi(p), p - u \rangle \geq 0 \qquad \forall u \in \mathcal{C}.$$

By Lemma D.1,

$$D_\psi(u, s_0) - D_\psi(u, p) - D_\psi(p, s_0) = \langle \nabla\psi(s_0) - \nabla\psi(p), p - u \rangle .$$

Thus Part 2 follows.

Assume Part 2. The same identity implies

$$\langle \nabla\psi(s_0) - \nabla\psi(p), p - u \rangle \geq 0 \qquad \forall u \in \mathcal{C}.$$

Hence

$$\langle \nabla\psi(p) - \nabla\psi(s_0), u - p \rangle \geq 0 \qquad \forall u \in \mathcal{C}.$$

This is the variational inequality for minimizing $D_\psi(s, s_0)$ over $\mathcal{C}$. Strict convexity gives uniqueness, so Part 1 follows. □

Corollary D.3 (Fejer monotonicity and bounded drift)**.** *Let* $s_{t+1} = \operatorname{argmin}_{s \in \mathcal{C}_t} D_\psi(s, s_t)$. *If* $u^\star \in \cap_{t=1}^T \mathcal{C}_t$, *then*

$$D_\psi(u^\star, s_{t+1}) \leq D_\psi(u^\star, s_t)$$

and

$$\sum_{t=1}^T D_\psi(s_{t+1}, s_t) \leq D_\psi(u^\star, s_1).$$

Proof. By Theorem D.2,

$$D_\psi(u^\star, s_{t+1}) + D_\psi(s_{t+1}, s_t) \leq D_\psi(u^\star, s_t).$$

Dropping the second term gives the first claim. Summing gives

$$\begin{aligned}
\sum_{t=1}^T D_\psi(s_{t+1}, s_t) &\leq \sum_{t=1}^T \left(D_\psi(u^\star, s_t) - D_\psi(u^\star, s_{t+1}) \right) \\
&= D_\psi(u^\star, s_1) - D_\psi(u^\star, s_{T+1}) \\
&\leq D_\psi(u^\star, s_1).
\end{aligned}$$

□

Theorem D.4 (Soft projection inequality)**.** *Let $\ell_t : S \to \mathbb{R}$ be convex and let*

$$s_{t+1} = \operatorname*{argmin}_{s \in S} \{D_\psi(s, s_t) + \eta_t \ell_t(s)\}.$$

Then for all $u \in S$,

$$\eta_t(\ell_t(s_{t+1}) - \ell_t(u)) \leq D_\psi(u, s_t) - D_\psi(u, s_{t+1}) - D_\psi(s_{t+1}, s_t).$$

Proof. For some $g_{t+1} \in \partial \ell_t(s_{t+1})$, optimality gives

$$\langle \nabla\psi(s_{t+1}) - \nabla\psi(s_t) + \eta_t g_{t+1}, u - s_{t+1} \rangle \geq 0.$$

Thus

$$\eta_t \langle g_{t+1}, s_{t+1} - u \rangle \leq \langle \nabla\psi(s_t) - \nabla\psi(s_{t+1}), s_{t+1} - u \rangle .$$

Convexity gives

$$\ell_t(s_{t+1}) - \ell_t(u) \leq \langle g_{t+1}, s_{t+1} - u \rangle .$$

Lemma D.1 gives

$$\langle \nabla\psi(s_t) - \nabla\psi(s_{t+1}), s_{t+1} - u \rangle = D_\psi(u, s_t) - D_\psi(u, s_{t+1}) - D_\psi(s_{t+1}, s_t).$$

Combining proves the claim. □

Theorem D.5 (Soft projection as implicit mirror descent, formal version of Theorem 3.2)**.** *Let ℓ_t be differentiable and let*

$$s_{t+1} = \operatorname*{argmin}_{s} \{D_\psi(s, s_t) + \eta \ell_t(s)\}.$$

If s_{t+1} is an interior optimum, then

$$\nabla\psi(s_{t+1}) = \nabla\psi(s_t) - \eta \nabla \ell_t(s_{t+1}).$$

If $\psi(s) = \|s\|_2^2 / 2$, then

$$s_{t+1} = s_t - \eta \nabla \ell_t(s_{t+1}).$$

Proof. At an interior optimum,

$$0 = \nabla_s D_\psi(s_{t+1}, s_t) + \eta \nabla \ell_t(s_{t+1}).$$

Since

$$\nabla_s D_\psi(s, s_t) = \nabla\psi(s) - \nabla\psi(s_t),$$

we obtain

$$\nabla\psi(s_{t+1}) - \nabla\psi(s_t) + \eta \nabla \ell_t(s_{t+1}) = 0.$$

Rearranging proves the mirror form. If $\psi(s) = \|s\|_2^2 / 2$, then $\nabla\psi(s) = s$. □

Theorem D.6 (Quadratic projection memory as implicit gradient descent, formal version of Theorem 3.3)**.** *Let* $\ell(W) = \|Wk - v\|^2/2$*. The update*

$$W_{t+1} = \operatorname*{argmin}_W \left\{ \frac{1}{2} \|W - W_t\|_F^2 + \eta \ell(W) \right\}$$

is equivalent to

$$W_{t+1} = W_t - \eta \nabla \ell(W_{t+1})$$

and has closed form

$$W_{t+1} = W_t + \frac{\eta}{1 + \eta \|k\|^2} (v - W_t k) k^\top.$$

Moreover, if $e = v - W_t k$*, then*

$$v - W_{t+1} k = \frac{1}{1 + \eta \|k\|^2} e.$$

Proof. The gradient is

$$\nabla \ell(W) = (Wk - v) k^\top.$$

The optimality condition for the proximal objective is

$$0 = W_{t+1} - W_t + \eta (W_{t+1} k - v) k^\top,$$

which is the implicit gradient step. Let $e = v - W_t k$ and set $W_{t+1} = W_t + \alpha e k^\top$. Then

$$v - W_{t+1} k = e - \alpha e \|k\|^2 = (1 - \alpha \|k\|^2) e.$$

The implicit step gives

$$\alpha e k^\top = \eta (1 - \alpha \|k\|^2) e k^\top.$$

Thus

$$\alpha = \frac{\eta}{1 + \eta \|k\|^2}.$$

Substituting α gives the update and residual contraction. □

Theorem D.7 (KL consistency projection)**.** *Let* p_t *be strictly positive on a finite space* $\mathcal{X}$*. The solution of*

$$\operatorname*{argmin}_{p : \mathbb{E}_p[\phi] = c} D_{\mathrm{KL}}(p \| p_t)$$

has the form

$$p_{t+1}(x) = \frac{p_t(x) \exp(\lambda^\top \phi(x))}{Z(\lambda)}.$$

Proof. The Lagrangian is

$$\mathcal{L}(p, \lambda, \nu) = \sum_x p(x) \log \frac{p(x)}{p_t(x)} + \lambda^\top \left(\sum_x p(x)\phi(x) - c \right) + \nu \left(\sum_x p(x) - 1 \right).$$

Stationarity gives

$$\log \frac{p(x)}{p_t(x)} + 1 + \lambda^\top \phi(x) + \nu = 0.$$

Hence, after absorbing signs into λ and normalizing,

$$p_{t+1}(x) = \frac{p_t(x) \exp(\lambda^\top \phi(x))}{Z(\lambda)}.$$

□

Proposition D.8 (Gradient descent as linearized projection)**.** *For $D_\psi(s, s_t) = \|s - s_t\|_2^2/2$, replacing $\ell_t(s)$ in the soft projection objective by its first-order approximation at s_t gives $s_{t+1} = s_t - \eta \nabla \ell_t(s_t)$.*

Proof. The linearized objective is

$$\frac{1}{2} \|s - s_t\|^2 + \eta \langle \nabla \ell_t(s_t), s - s_t \rangle + \eta \ell_t(s_t).$$

The stationarity condition is

$$s - s_t + \eta \nabla \ell_t(s_t) = 0.$$

Thus

$$s_{t+1} = s_t - \eta \nabla \ell_t(s_t).$$

□

E Conservative Memory Proofs

Theorem E.1 (Exact associative binding, formal version of Theorem 3.4)**.** *Let $K \in \mathbb{R}^{r \times T}$ and $V \in \mathbb{R}^{m \times T}$. The set $\mathcal{C} = \{W \in \mathbb{R}^{m \times r} : WK = V\}$ is nonempty if and only if $\ker K \subseteq \ker V$. If $\mathcal{C}$ is nonempty, then*

$$\operatorname*{argmin}_{W \in \mathcal{C}} \frac{1}{2} \|W - W_0\|_F^2 = W_0 + (V - W_0 K) K^+.$$

Proof. If $WK = V$ and $a \in \ker K$, then

$$Va = WKa = 0.$$

Thus $\ker K \subseteq \ker V$.

Assume $\ker K \subseteq \ker V$. Since $K^+ K$ is the orthogonal projector onto $(\ker K)^\perp$, the condition implies

$$V = VK^+K.$$

Thus $W = VK^+$ satisfies $WK = V$.

Let $R = V - W_0K$ and $W = W_0 + \Delta$. The projection problem is

$$\min_{\Delta} \frac{1}{2} \|\Delta\|_F^2 \quad \text{subject to} \quad \Delta K = R.$$

The minimum-norm solution is

$$\Delta^\star = RK^+.$$

Therefore

$$W^\star = W_0 + (V - W_0K)K^+.$$

□

Theorem E.2 (Innovation decomposition, formal version of Theorem 3.5)**.** *Assume $WK = V$. Let $\Pi = KK^+$, $z = (I - \Pi)k$, and $a = K^+k$. Then*

$$Wk = Va + Wz.$$

If $z = 0$, Wk is fixed over all old-feasible W. If $z \neq 0$, the constraint $Wk = v$ is equivalent over old-feasible W to $Wz = v - Va$.

Proof. The decomposition is

$$k = KK^+k + (I - KK^+)k = Ka + z.$$

Multiplying by W gives

$$Wk = WKa + Wz = Va + Wz.$$

If $z = 0$, then $Wk = Va$ for every W satisfying $WK = V$. If $z \neq 0$, then

$$Wk = v \quad \Longleftrightarrow \quad Va + Wz = v \quad \Longleftrightarrow \quad Wz = v - Va.$$

□

Theorem E.3 (Conflict impossibility, formal version of Theorem 3.6)**.** *Assume $W_tK = V$ and $z = (I - KK^+)k = 0$. If $W_t + \Delta$ preserves old evidence, then $(W_t + \Delta)k = W_tk$. Hence $Wk = v$ is compatible with old evidence if and only if $v = W_tk$.*

Proof. Preserving old evidence means

$$(W_t + \Delta)K = V = W_tK.$$

Thus

$$\Delta K = 0.$$

If $z = 0$, then $k \in \operatorname{span}(K)$, hence

$$\Delta k = 0.$$

Therefore

$$(W_t + \Delta)k = W_tk.$$

□

Theorem E.4 (Hard Conservative Projection Memory)**.** *Assume $W_t K = V$, let $\Pi = KK^+$, $z = (I - \Pi)k \neq 0$, and $e = v - W_t k$. Then*

$$W_{t+1} = W_t + \frac{ez^\top}{\|z\|^2}$$

preserves $WK = V$, satisfies $Wk = v$, and minimizes $\|W - W_t\|_F^2 / 2$ among all memories satisfying both constraints.

Proof. Let $\Delta^\star = ez^\top / \|z\|^2$. Since $z \perp \operatorname{span}(K)$,

$$\Delta^\star K = 0.$$

Thus

$$W_{t+1} K = W_t K = V.$$

Also $z^\top k = \|z\|^2$, so

$$\Delta^\star k = e.$$

Hence

$$W_{t+1} k = W_t k + e = v.$$

For minimality, any feasible update Δ satisfies

$$\Delta K = 0, \qquad \Delta k = e.$$

Since $\Delta K = 0$,

$$\Delta k = \Delta(I - \Pi)k = \Delta z.$$

Thus the problem reduces to

$$\min_{\Delta} \frac{1}{2} \|\Delta\|_F^2 \quad \text{subject to} \quad \Delta z = e.$$

The minimum-norm solution is $\Delta^\star = ez^\top / \|z\|^2$. □

Theorem E.5 (No-free-repair lower bound, formal version of Theorem 3.8)**.** *Assume $W_t K = V$, let $\Pi = KK^+$, $z = (I - \Pi)k$, and $e = v - W_t k$. Let $W_t + \Delta$ preserve old constraints, so $\Delta K = 0$. If $z \neq 0$ and*

$$\|v - (W_t + \Delta)k\| \leq \gamma \|e\|$$

for $0 \leq \gamma < 1$, then

$$\|\Delta\|_F \geq \frac{(1 - \gamma)\|e\|}{\|z\|}.$$

If $z = 0$ and $e \neq 0$, no update preserving old constraints can achieve such a contraction.

Proof. Since $\Delta K = 0$,

$$\Delta k = \Delta(I - \Pi)k = \Delta z.$$

Let

$$e^+ = v - (W_t + \Delta)k.$$

Then

$$e^+ = e - \Delta z.$$

Thus

$$\|\Delta z\| = \left\|e - e^+\right\| \geq \|e\| - \left\|e^+\right\| \geq (1 - \gamma)\|e\|.$$

Also

$$\|\Delta z\| \leq \|\Delta\|_F \|z\|.$$

If $z \neq 0$, division gives the result. If $z = 0$, then $\Delta z = 0$ and $e^+ = e$, so no contraction with $\gamma < 1$ is possible when $e \neq 0$. □

Theorem E.6 (Optimal contraction, formal version of Theorem 3.9)**.** *Assume $z \neq 0$ and fix $0 \leq \gamma \leq 1$. Among all updates satisfying*

$$\Delta K = 0, \qquad v - (W_t + \Delta)k = \gamma e,$$

the unique minimum-Frobenius-norm update is

$$\Delta^\star = (1 - \gamma)\frac{ez^\top}{\|z\|^2}.$$

Its norm is

$$\|\Delta^\star\|_F = \frac{(1 - \gamma)\|e\|}{\|z\|}.$$

Proof. The residual constraint is

$$e - \Delta k = \gamma e.$$

Since $\Delta K = 0$, $\Delta k = \Delta z$. Thus

$$\Delta z = (1 - \gamma)e.$$

The minimum-norm solution of this linear constraint is

$$\Delta^\star = (1 - \gamma)\frac{ez^\top}{\|z\|^2}.$$

It preserves old constraints because $z^\top K = 0$:

$$\Delta^\star K = 0.$$

It achieves the target because

$$\Delta^\star z = (1-\gamma)e.$$

The norm is

$$\|\Delta^\star\|_F = (1-\gamma)\frac{\|e\|\,\|z\|}{\|z\|^2} = \frac{(1-\gamma)\,\|e\|}{\|z\|}.$$

□

Theorem E.7 (Sequential hard CPM equals batch projection, formal version of Theorem 3.10)**.** *Let $(k_t, v_t)_{t=1}^T$ be compatible evidence, and let $K_t = [k_1, \dots, k_t]$ and $V_t = [v_1, \dots, v_t]$. Initialize $W_1 = W_0$. At step t, let $\Pi_{t-1} = K_{t-1}K_{t-1}^+$ and $z_t = (I - \Pi_{t-1})k_t$. If $z_t = 0$, make no update. If $z_t \neq 0$, apply hard CPM:*

$$W_{t+1} = W_t + \frac{(v_t - W_t k_t)z_t^\top}{\|z_t\|^2}.$$

Then

$$W_{T+1} = \operatorname*{argmin}_{W:WK_T=V_T} \frac{1}{2}\|W - W_0\|_F^2.$$

Proof. Let

$$\mathcal{C}_t = \{W : WK_t = V_t\}, \qquad \mathcal{T}_t = \{H : HK_t = 0\}.$$

We prove by induction that W_{t+1} is the Euclidean projection of W_0 onto $\mathcal{C}_t$. For $t = 0$, this is $W_1 = W_0$.

Assume $W_t = \mathrm{Proj}_{\mathcal{C}_{t-1}}(W_0)$. Thus $W_t \in \mathcal{C}_{t-1}$ and

$$\langle H, W_t - W_0\rangle_F = 0 \qquad \forall H \in \mathcal{T}_{t-1}.$$

If $z_t = 0$, compatibility implies the new constraint is already implied by $\mathcal{C}_{t-1}$, hence $\mathcal{C}_t = \mathcal{C}_{t-1}$ and no update preserves the projection.

If $z_t \neq 0$, hard CPM gives $W_{t+1} \in \mathcal{C}_t$ by Theorem E.4. Let $H \in \mathcal{T}_t$. Then $H \in \mathcal{T}_{t-1}$ and $Hk_t = 0$. Since $\Pi_{t-1}k_t \in \mathrm{span}(K_{t-1})$,

$$Hz_t = Hk_t - H\Pi_{t-1}k_t = 0.$$

Let $\Delta_t = W_{t+1} - W_t$. Then

$$\langle H, \Delta_t\rangle_F = \left\langle H, (v_t - W_t k_t)z_t^\top\right\rangle_F \frac{1}{\|z_t\|^2} = \frac{(v_t - W_t k_t)^\top H z_t}{\|z_t\|^2} = 0.$$

Therefore

$$\langle H, W_{t+1} - W_0\rangle_F = \langle H, W_t - W_0\rangle_F + \langle H, \Delta_t\rangle_F = 0.$$

The Euclidean projection characterization onto the affine set $\mathcal{C}_t$ gives $W_{t+1} = \mathrm{Proj}_{\mathcal{C}_t}(W_0)$. □

Theorem E.8 (Soft Conservative Projection Memory, formal version of Theorem 3.7)**.** *Assume $W_t K = V$, let $\Pi = KK^+$, $z = (I - \Pi)k \neq 0$, and $e = v - W_t k$. The unique solution of*

$$\operatorname*{argmin}_{W:WK=V} \left\{ \frac{1}{2} \|W - W_t\|_F^2 + \frac{\eta}{2} \|Wk - v\|^2 \right\}$$

is

$$W_{t+1} = W_t + \frac{\eta}{1 + \eta \|z\|^2} e z^\top.$$

Moreover,

$$W_{t+1} K = V, \qquad v - W_{t+1} k = \frac{1}{1 + \eta \|z\|^2} e.$$

Proof. Write $W = W_t + \Delta$. The constraint $WK = V$ is equivalent to

$$\Delta K = 0.$$

Since $\Delta K = 0$,

$$\Delta k = \Delta (I - \Pi) k = \Delta z.$$

The objective becomes

$$\min_{\Delta:\Delta K=0} \frac{1}{2} \|\Delta\|_F^2 + \frac{\eta}{2} \|\Delta z - e\|^2 .$$

The stationarity equation in the feasible subspace is

$$\Delta + \eta(\Delta z - e) z^\top = 0.$$

Let $\Delta = \alpha e z^\top$. Substitution gives

$$\alpha e z^\top + \eta(\alpha e \|z\|^2 - e) z^\top = 0.$$

Thus

$$\alpha(1 + \eta \|z\|^2) = \eta.$$

Therefore

$$\Delta = \frac{\eta}{1 + \eta \|z\|^2} e z^\top.$$

Since $z^\top K = 0$, $W_{t+1} K = V$. The residual is

$$\begin{aligned} v - W_{t+1} k &= e - \Delta k \\ &= e - \Delta z \\ &= e - \frac{\eta \|z\|^2}{1 + \eta \|z\|^2} e \\ &= \frac{1}{1 + \eta \|z\|^2} e. \end{aligned}$$

□

F Ridge and Bayesian Projection Memory Proofs

Theorem F.1 (Ridge solution)**.** *Let*

$$J_t(W) = \frac{\lambda}{2} \|W - W_0\|_F^2 + \frac{1}{2} \sum_{\tau=1}^{t} \rho_\tau \|Wk_\tau - v_\tau\|^2 .$$

Define

$$A_t = \lambda I + \sum_{\tau=1}^{t} \rho_\tau k_\tau k_\tau^\top, \qquad B_t = \lambda W_0 + \sum_{\tau=1}^{t} \rho_\tau v_\tau k_\tau^\top.$$

Then the unique minimizer is $W_{t+1} = B_t A_t^{-1}$.

Proof. Differentiate:

$$\nabla J_t(W) = \lambda(W - W_0) + \sum_{\tau=1}^{t} \rho_\tau (Wk_\tau - v_\tau) k_\tau^\top.$$

At optimum,

$$0 = \lambda(W - W_0) + \sum_{\tau=1}^{t} \rho_\tau (Wk_\tau - v_\tau) k_\tau^\top.$$

Thus

$$W \left(\lambda I + \sum_{\tau=1}^{t} \rho_\tau k_\tau k_\tau^\top \right) = \lambda W_0 + \sum_{\tau=1}^{t} \rho_\tau v_\tau k_\tau^\top.$$

Since $\lambda > 0$, A_t is invertible. Hence

$$W_{t+1} = B_t A_t^{-1}.$$

□

Theorem F.2 (Recursive ridge update)**.** *Let* $P_t = A_{t-1}^{-1}$ *and* $W_t = B_{t-1} P_t$. *Define*

$$g_t = \frac{\rho_t P_t k_t}{1 + \rho_t k_t^\top P_t k_t}.$$

Then

$$W_{t+1} = W_t + (v_t - W_t k_t) g_t^\top, \qquad P_{t+1} = P_t - \frac{\rho_t P_t k_t k_t^\top P_t}{1 + \rho_t k_t^\top P_t k_t}.$$

Proof. The Sherman-Morrison identity gives

$$P_{t+1} = A_t^{-1} = P_t - \frac{\rho_t P_t k_t k_t^\top P_t}{1 + \rho_t k_t^\top P_t k_t}.$$

Let $a_t = k_t^\top P_t k_t$. Then

$$k_t^\top P_{t+1} = k_t^\top P_t - \frac{\rho_t a_t k_t^\top P_t}{1+\rho_t a_t} = \frac{k_t^\top P_t}{1+\rho_t a_t}.$$

Using $B_t = B_{t-1} + \rho_t v_t k_t^\top$,

$$\begin{aligned} W_{t+1} &= B_t P_{t+1} \\ &= B_{t-1} P_{t+1} + \rho_t v_t k_t^\top P_{t+1} \\ &= W_t - \frac{\rho_t W_t k_t k_t^\top P_t}{1+\rho_t a_t} + \frac{\rho_t v_t k_t^\top P_t}{1+\rho_t a_t} \\ &= W_t + (v_t - W_t k_t)\frac{\rho_t k_t^\top P_t}{1+\rho_t a_t} \\ &= W_t + (v_t - W_t k_t) g_t^\top. \end{aligned}$$

□

Theorem F.3 (Ridge bound, formal version of Theorem 3.12)**.** *Let*

$$W_t = \operatorname*{argmin}_W J_{t-1}(W), \qquad P_t = A_{t-1}^{-1}, \qquad a_t = k_t^\top P_t k_t, \qquad e_t = v_t - W_t k_t.$$

For every $U \in \mathbb{R}^{m\times r}$,

$$\sum_{t=1}^{T} \frac{\rho_t}{1+\rho_t a_t} \|e_t\|^2 \le \lambda \|U - W_0\|_F^2 + \sum_{t=1}^{T} \rho_t \|v_t - U k_t\|^2 .$$

Proof. Since W_t minimizes J_{t-1}, for any Δ,

$$J_{t-1}(W_t + \Delta) = J_{t-1}(W_t) + \frac{1}{2}\operatorname{tr}(\Delta A_{t-1} \Delta^\top).$$

Also

$$J_t(W_t + \Delta) = J_{t-1}(W_t + \Delta) + \frac{\rho_t}{2}\|(W_t + \Delta)k_t - v_t\|^2 .$$

With $e_t = v_t - W_t k_t$, the increment minimization is

$$\min_\Delta \left\{ \frac{1}{2}\operatorname{tr}(\Delta A_{t-1}\Delta^\top) + \frac{\rho_t}{2}\|\Delta k_t - e_t\|^2 \right\}.$$

Solving row-wise gives minimum value

$$\frac{\rho_t}{2(1+\rho_t a_t)}\|e_t\|^2 .$$

Hence

$$J_t(W_{t+1}) - J_{t-1}(W_t) = \frac{\rho_t}{2(1+\rho_t a_t)}\|e_t\|^2 .$$

Summing over t and using $W_1 = W_0$ gives

$$\sum_{t=1}^{T} \frac{\rho_t}{2(1+\rho_t a_t)}\|e_t\|^2 = J_T(W_{T+1}).$$

Since W_{T+1} minimizes J_T,

$$J_T(W_{T+1}) \leq J_T(U) = \frac{\lambda}{2} \|U - W_0\|_F^2 + \frac{1}{2} \sum_{t=1}^{T} \rho_t \|Uk_t - v_t\|^2 .$$

Multiplying by 2 proves the claim. □

Theorem F.4 (Log-determinant information identity, formal version of Theorem 3.13)**.** *Let* $A_t = \lambda I + \sum_{\tau=1}^{t} \rho_\tau k_\tau k_\tau^\top$ *and* $a_t = k_t^\top A_{t-1}^{-1} k_t$*. Then*

$$\sum_{t=1}^{T} \log(1 + \rho_t a_t) = \log \frac{\det A_T}{\det A_0}.$$

Proof. The matrix determinant lemma gives

$$\begin{aligned} \det A_t &= \det(A_{t-1} + \rho_t k_t k_t^\top) \\ &= \det A_{t-1}(1 + \rho_t k_t^\top A_{t-1}^{-1} k_t) \\ &= \det A_{t-1}(1 + \rho_t a_t). \end{aligned}$$

Taking logarithms,

$$\log \det A_t - \log \det A_{t-1} = \log(1 + \rho_t a_t).$$

Summing over t gives the identity. □

Theorem F.5 (Bayesian ridge projection memory, formal version of Theorem 3.11)**.** *Assume each row* w_j *of* W *has prior* $w_j \sim \mathcal{N}(w_{0,j}, \lambda^{-1} I)$ *and observations satisfy* $v_t = W k_t + \varepsilon_t$ *with* $\varepsilon_t \sim \mathcal{N}(0, \rho_t^{-1} I_m)$ *independently over time. Then the posterior mean and MAP estimate are* $B_t A_t^{-1}$*.*

Proof. The negative log posterior, up to constants independent of W, is

$$\frac{\lambda}{2} \|W - W_0\|_F^2 + \frac{1}{2} \sum_{\tau=1}^{t} \rho_\tau \|W k_\tau - v_\tau\|^2 .$$

This is $J_t(W)$. The posterior is Gaussian because the prior is Gaussian and the observation model is linear Gaussian. Hence the posterior mean equals the MAP. By Theorem F.1, this value is

$$B_t A_t^{-1}.$$

□